

# BACSWN VnV and ICD Plan

## BACSWN — Master Verification & Validation (V&V) and Interface Control (ICD) Plan

**Document:** BACSWN-VV-2026-006 **Classification:** CONFIDENTIAL — For Authorized Recipients Only **Prepared for:** Owner & CEO, BACSWN (private delivery partner) **Prepared by:** Sky Miles Limited — AI Elevate Division **Version:** 1.0 — June 2026 **Companion to:** Mesh Architecture Master Plan (BACSWN-MARCH-2026-002); Master Framework Schedule (BACSWN-SCHED-2026-003)

### 1. Purpose

This plan defines **how BACSWN is proven correct, fit-for-purpose, and interoperable** — the verification & validation (V&V) strategy across every test level, and the interface control (ICD) regime that keeps independently-built components and external systems working together. It is the authoritative reference for test planning, acceptance, and interface change control, and it underpins the gate exit-criteria in the Master Framework Schedule.

- Verification** — "did we build it right?" (against the architecture and requirements).
- Validation** — "did we build the right thing?" (it works in the real Bahamian operational world).

### 2. V&V Principles

- Requirements-traceable:** every requirement maps to at least one test via a **Requirements Traceability Matrix (RTM)**; no requirement ships unverified.
- Gated:** each schedule gate has defined **entry/exit criteria**; a gate cannot close on an open critical defect.
- Independent where it matters:** security and survivability use **independent V&V (IV&V)** — third-party penetration testing and independent structural certification.
- Test like you fly:** validation includes a **live hurricane-season** exercise and real inter-agency drills, not just lab conditions.
- Regression-protected:** automated regression guards the live platform (already 137/137 tests, 15/15 security) as the network grows.

### 3. V&V Levels & Test Stages

| Level                         | What it proves                                                                           | Ties to             |
|-------------------------------|------------------------------------------------------------------------------------------|---------------------|
| Unit / component              | Modules, services, message handlers                                                      | CI regression suite |
| Factory Acceptance Test (FAT) | Edge node, radios, sensors, <b>Raytheon radar</b> accepted on the bench before shipment  | G3/G4, RAD-3        |
| Integration test              | Mesh protocol, routing, store-and-forward, consensus, hub coordination                   | G5–G8               |
| Site Acceptance Test (SAT)    | Per-station commissioning against the §11 acceptance criteria                            | G5/G9/G10/G11       |
| System test                   | End-to-end across the full 15-node network + radar + IoT                                 | G11                 |
| Security V&V (IV&V)           | Crypto, secure boot, signed-OTA, tamper-zeroization, penetration test                    | SEC-1/2/3           |
| Survivability V&V             | Structural cert, missile-impact, lightning/grounding, power-cut & autonomous-mode drills | STR-1, G7           |
| Operational validation        | Live hurricane-season soak, inter-agency activation drill, hub failover                  | G13, EMG-3, DR-1    |

**Per-station SAT (from Architecture §11)** — a station is accepted only when all transport layers pass link-budget + live-traffic tests; it sustains its rated battery autonomy; joins the mesh and a consensus vote; runs edge inference within latency; passes an autonomous-mode drill; completes an OTA A/B update; is Cat 4/5 / lightning / corrosion certified; and has verified cryptographic provisioning.

### 4. Test Management

- Requirements Traceability Matrix (RTM)** — living artifact linking requirement → design → test → result → defect.
- Defect management** — severity-classified; **no gate closes on an open critical/major defect**; trend reviewed weekly with the owner/CEO.

- **Entry/exit criteria** — documented per gate; signed off by the program technical authority.
- **Test environments** — CI/lab bench → staging → **pilot cluster** (MYNN/MYGF/MYAM/MYEG) → production.
- **Evidence** — every gate produces a test report and traceability evidence retained as program record.

## 5. Interface Control (ICD) Regime

Every interface — internal and external — is governed by a versioned **Interface Control Document** specifying: protocol, data schema/message types, **security binding (Architecture §4.4)**, versioning & backward-compatibility rules, error handling, throughput/SLA, ownership, and change control. ICDs are baselined early (gate VV-1) and held under change control for the life of the program.

### 5.1 Internal Interfaces

| ICD    | Interface                                                                                 |
|--------|-------------------------------------------------------------------------------------------|
| ICD-01 | Edge node ↔ mesh protocol (message schema, routing, store-and-forward) — versioned per P6 |
| ICD-02 | Edge node ↔ hub control plane (HEARTBEAT/OBSERVATION, OTA, topology)                      |
| ICD-03 | Sensor & radio HAL ↔ node (sensor suite + VHF/cellular/satellite radios)                  |
| ICD-04 | IoT (LoRaWAN/NB-IoT) sensors ↔ station gateway                                            |
| ICD-05 | <b>Raytheon primary radar</b> ↔ surveillance fusion                                       |
| ICD-06 | Drone/UAS ↔ COP / dispatch / telemetry                                                    |
| ICD-07 | Hub ↔ standby hub (replication, failover)                                                 |

### 5.2 External Interfaces

| ICD    | Interface                                                                          |
|--------|------------------------------------------------------------------------------------|
| ICD-10 | Emergency agencies (NEMA, Police, Fire/Rescue, RBDF) — <b>CAP / EDXL</b> + COP API |
| ICD-11 | Emergency shelter system (activation, capacity/occupancy, evacuation routing)      |
| ICD-12 | Overflight/landing-fee accounting ↔ civil-aviation authority billing               |
| ICD-13 | Weather feeds — AWC, NWS, OpenMeteo, <b>Tomorrow.io</b> (space-based radar)        |
| ICD-14 | ICAO <b>OPMET / WAFS / AFTN-AMHS</b> aviation-weather exchange                     |
| ICD-15 | United States (phased) — NOAA/NWS/NHC, FAA Miami Oceanic (ZMA), <b>NORAD</b>       |

### 5.3 ICD Governance

- **ICD register** maintained by the integration lead; each ICD versioned and owned.
- **Backward-compatible evolution** preferred; breaking changes require explicit versioning, notice, and coordinated client updates (we have live external integrations and do not break them casually).
- External ICDs (agency/international/defence) are agreed bilaterally and inherit the data-governance sharing rules (BACSWN-GOV-2026-005).

## 6. V&V / ICD Milestones (map to Schedule)

| Gate                                                                                      | Month | Meaning                                                                                                                                   |
|-------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VV-1</b>                                                                               | 3     | Master V&V plan + <b>ICD register baselined</b> ; RTM established                                                                         |
| (FAT/SAT/integration run continuously against core gates G4–G11, SEC-1/2, RAD-3, EMG-1/2) |       |                                                                                                                                           |
| <b>VV-2</b>                                                                               | 23    | <b>Operational validation complete</b> — live hurricane-season exercise + full inter-agency drill + hub failover passed; RTM fully closed |

See also: *Master Framework Schedule* (gates VV-1, VV-2, and the per-capability gates they verify); *Architecture §11* (per-station acceptance); *Data Governance & Sovereignty Policy* (BACSWN-GOV-2026-005).